Andrew W. Shenstone

Shenstone, Andrew - Application for Principal Engineer (UI Platform & AI-Native UX) @ Tripleseat

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SUMMARY

I am a senior front-end–leaning full-stack engineer with 12+ years shipping production UIs on SaaS and high-traffic products—including a Shadcn/Tailwind stack on a complex Next.js app, reusable component patterns, and AI features users touch every day (vision, chat, embeddings, background jobs). I have led adoption of a design system on a deadline (Chakra at Tilia), owned cohesive UI architecture on multi-surface products, and modernized front-end stacks where legacy and new had to coexist.

Tripleseat's mandate maps cleanly to how I like to work: own the component system on top of **shadcn/ui** and **Radix**, align tokens and accessibility baselines, and pair with design and product so AI surfaces—streaming, tooling, human-in-the-loop—feel first-class. You are on **React 18** today; moving to **React 19** is a natural part of that modernization, and I am comfortable planning and landing that upgrade as part of the broader **Kendo → shadcn** consolidation rather than treating it as a separate science project.

Hospitality-specific domain depth is not my headline credential—your bonus list calls that out—but B2B workflow software, busy operational users, and integrate-heavy products are familiar territory. What I have not shipped yet is a dedicated **UI MCP server** for an agentic internal toolchain; I would treat that as a focused collaboration with platform and architecture, alongside clear standards so every "AI bet" lands cleanly in the product.

SKILLS

UI platform & modern React: React 18 production depth; React 19 upgrades are a normal next step—not a blocker—for a consolidation effort like yours, TypeScript-first applications, Tailwind CSS; Shadcn UI in production (not prototype-only); Radix

primitives through shadcn and direct where needed, Design tokens, theming, composition patterns, accessibility-minded components, Performance budgets and rendering discipline on SaaS-scale UIs

AI-native UX & engineering workflows: Production AI-assisted UX: vision and language models, embeddings, RAG-style flows, conversational surfaces, Streaming and async UI patterns for model output, Daily use of Cursor-class tooling; comfortable championing AI-accelerated SDLC with guardrails, Partnering with product and design on patterns that feel native, not bolted on

Testing, migration & org-scale standards: Component testing and E2E (Vitest, Cypress); ready to help define visual regression as part of a single standard, Large front-end migrations without freezing product (e.g., spearheaded Chakra UI adoption on a constrained timeline), Legacy stack modernization experience (e.g., Webpack/build debt, dual-codebase agency contexts—not Kendo specifically, but the playbook transfers), Mentorship through pairing, code review, and crisp reviews with design leadership

Architecture & async: Monorepos, Async workflows (Inngest, webhooks, background jobs), ETL pipelines, Performance optimization

EXPERIENCE

Founder & Lead Engineer

Plantfolio | November 2023 - Present

Product Leadership & Architecture

- Architected and built a comprehensive plant management platform from the ground up, serving as both technical lead and product owner
- Owned the full feature cycle: joined customer calls to identify friction points, architected solutions, implemented, tested, and verified success in production
- Designed and implemented a sophisticated monorepo architecture with 12+ packages, enabling rapid development and maintainable codebases
- Led product strategy and user experience decisions, making product micro-decisions and navigating ambiguity to deliver value iteratively

Technical Achievements

- AI Integration: Built advanced AI-powered features including plant identification from photos, contextual chatbot (Sage), and smart text extraction from plant tags using OpenAI's vision and language models
- Modern Frontend Architecture: Implemented Next.js with React, React Server Components, and TypeScript, achieving exceptional performance and developer experience
- Complex Data Modeling: Designed and implemented sophisticated PostgreSQL schemas with Prisma, handling complex relationships between users, plants, folios, and media with 15+ interconnected models
- Supabase to Neon: Shipped on Supabase (PostgreSQL with Row-Level Security policies for multi-tenant isolation) before migrating the data layer to Neon; strong hands-on experience with RLS and Postgres-centric access control
- Content Management: Created sophisticated content management system with rich text editing, plant tagging, and flexible data structures

Scale & Performance

- Managed a database of 40,000+ plant species with advanced search capabilities using Algolia and vector embeddings
- Implemented comprehensive media management system with AWS S3, supporting image optimization, galleries, and metadata extraction
- Built robust authentication and authorization systems with Clerk, supporting multiple user types and subscription tiers
- Optimized application performance through React Server Components, efficient data fetching, and strategic caching

Client Case Study - AI Signage

- Delivered a production rollout for an independent garden center to automate plant signage generation; one independent retail customer in ongoing production who has been an enthusiastic reference
- Enabled bulk plant import (plain text) to printable signs in seconds; reduced a weekend-long manual process to minutes after initial setup

- Demonstrated ability to translate customer requirements into scalable technical solutions and shorten feedback loops

Full-Stack Ownership

- Developed complete payment integration with Stripe, subscription management, and user onboarding flows
- Built comprehensive email system with React Email, supporting transactional emails, notifications, and marketing campaigns
- Implemented advanced search and filtering capabilities with Algolia, supporting complex plant queries and user preferences

Code Quality & Leadership

- Established coding standards and architectural patterns that enabled rapid, consistent development
- Implemented comprehensive testing strategy with Vitest and Cypress, maintaining high code quality
- Built reusable component library and design system, ensuring consistent UI/UX across the platform

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Senior Software Engineer

Tilia | January 2023 - August 2023

- Partnered with cross-functional teams in product and design to revamp critical user-facing features
- Built and shipped work in support of KYC, identity verification, and risk-adjacent tooling as part of Tilia's payments platform (tools team context)
- Introduced and evolved a GraphQL API on the platform while consuming it from our product—hands-on with both server-side schema/resolvers and client integration
- Spearheaded and guided a team through the successful adoption of Chakra UI, reducing bespoke components and establishing cohesive design patterns

- Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
- Maintained exceptional unit test coverage of over 90% for the Tools team application, alongside comprehensive end-to-end testing
- Led the swift creation and deployment of a homepage reporting dashboard within the initial two months of joining
- Role ended in a company-wide layoff affecting approximately one-third of employees

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Senior Software Engineer

Bloom.wine | September 2020 - October 2022

- Architected a greenfield Next.js application for the primary admin interface of the Bloom product; much of that stack remains in production across many winery clients today
- Engineered Ruby on Rails GraphQL API endpoints shared by multiple applications across mobile and web devices
- CI/CD with CircleCI for build, test, and deployment workflows alongside the Next.js and Rails stack
- Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
- Collaborated with designers using Figma, establishing component library standards usable by both engineering and design
- Integrated DatoCMS and extended application functionality through third-party API integrations (Shopify, Mailchimp, Stripe, Shipstation, ActiveCampaign); all Bloom winery clients were on Shopify Plus

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Software Engineer II

Tune Inc - Marketplace Automation | August 2019 - May 2020

- Built and operated ETL-style data pipelines and integrations supporting marketplace automation workflows
- Automated a client workflow using a Node.js/Express API and React.js, reducing the time to complete the task from 10-15 minutes to 10-30 seconds
- Modernized the primary legacy React.js application by moving from a custom Webpack process to react-scripts, resulting in a 50% reduction in production build size and a 60% reduction in NPM dependencies
- Reduced development machine CPU usage from 75% to <5% by simplifying the build pipeline and dependency graph
- Role ended in a company-wide layoff

Software Developer

Ply Interactive Inc | June 2014 - July 2019

React Native

- Shipped multiple production React Native applications for agency clients—cross-platform mobile apps delivered end to end
- Washington State Fair: React Native app in sustained multi-year production; very high traffic during fair season, from tens of thousands of users into the hundreds of thousands cumulatively over seasons

Web & platforms

- Developed numerous web applications including Georgetownbeer.com, WashingtonWine.org, TwoWolvesWine.com, Novaerus.com, WellAir, and Plasma-air.com
- Built a standalone Ruby gem Content Management System, enabling simplified content management within Ruby on Rails projects
- Transitioned the agency to modern technologies like GraphQL, Gatsby.js, and React.js, improving delivery times for new projects
- Bloom.wine Product Development: Built extensive reporting features using complex PostgreSQL queries and interactive filtering, saving clients thousands of

hours on previously manual tasks

- Engineered a flexible Wine Club membership structure for wineries, allowing for diverse client configuration and broadening the range of potential sales targets
- Communicated and enforced the standards of Shopify's Polaris React.js component library with designers, resulting in efficiencies gained from leveraging pre-built components

COMPANY REFERENCE

Plantfolio (2023-Present) - AI-powered plant management platform with 40,000+ plant database serving both consumer garden management and independent garden center services - plantfolio.garden/

Tilia (2023) - All-in-one payment platform for digital economies - tilia.io/

Bloom.wine (2020-2022) - Commerce services for the alcohol industry; all clients on Shopify Plus - bloom.wine/

Tune Inc - Marketplace Automation (2019-2020) - SaaS platform for managing marketing partnerships across mobile and web - tune.com/

Ply Interactive Inc (2014-2019) - Boutique agency: web and production React Native apps (e.g. Washington State Fair at major scale); developed Bloom.wine product internally (2017-2019)